

Diagnosing Conformal Prediction Failures Under Distribution Shift

A COVID-19 Case Study

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SHAP Concentration Predicts GBM CP Failure

Pre-deployment risk signal for fragile GBM conformal predictors.

$\rho = 0.853$
primary correlation, $n = 16$

40%
exploratory cutoff

91.7%
SALT+WILDS check

Problem & Study Setting

Conformal Prediction (CP) gives uncertainty sets with coverage guarantees. Under deployment shift, those guarantees can break.

Research Question:

Can we **predict** when CP fails **before** deployment?

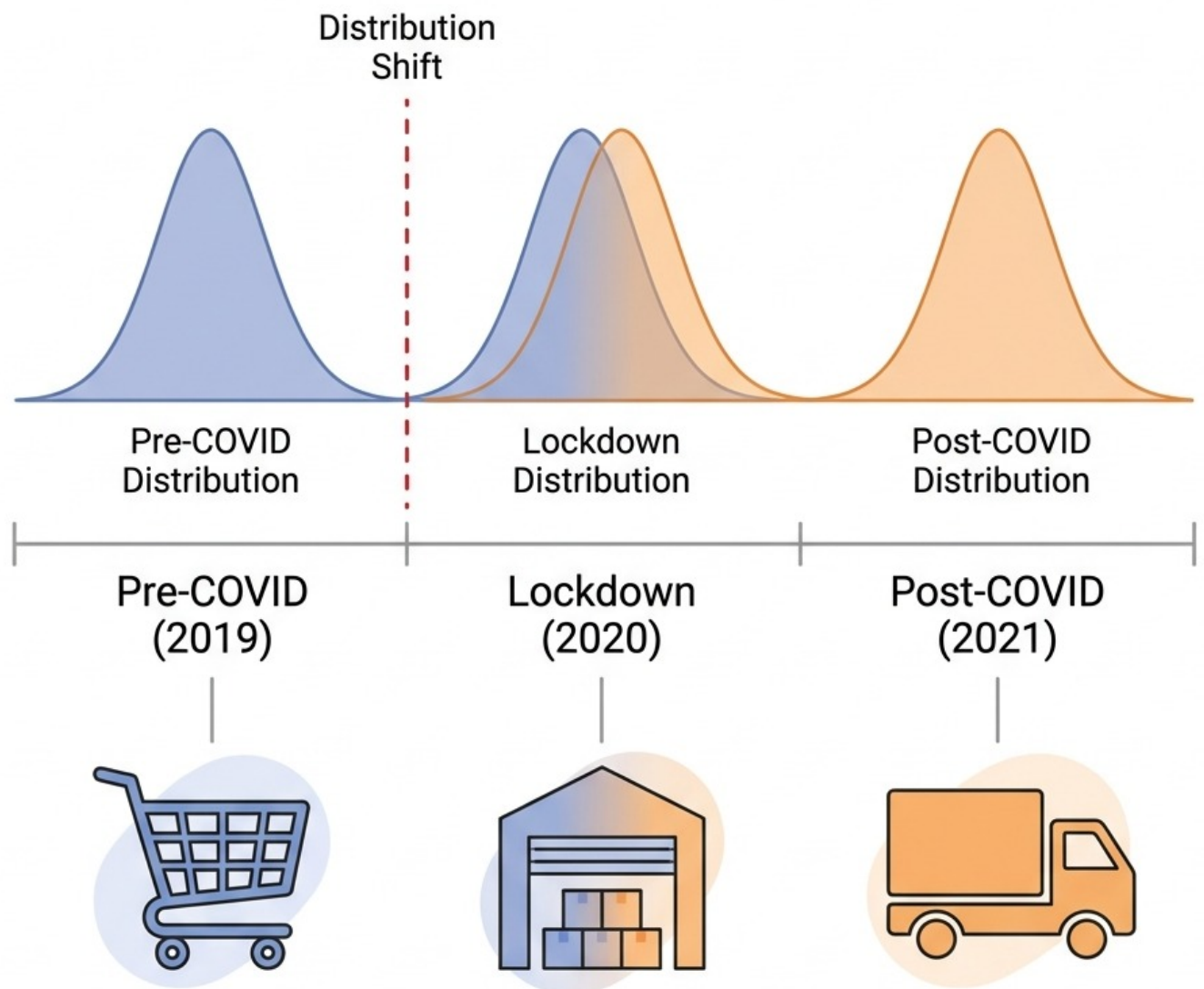


Fig 1: CP is calibrated pre-COVID, then tested after behavior changes.

Analysis set
SALT COVID shift: **8 tasks**
External multiclass endpoint: **+8 tasks**
Primary correlation: $n = 16$ **across 9 domains**

Diagnostic & Theorem

SHAP Concentration

$$C = \frac{\phi_1}{\sum_{j=1}^p \phi_j}$$

ϕ_j : mean absolute SHAP importance; ϕ_1 : top-ranked feature

Theorem (idealized additive model): concentration worsens APS score and coverage bounds

$$\mathbb{E}[s_{\text{test}}] \geq C(1 - (K - 1)\epsilon) + (1 - C)(1 - (K - 1)\bar{h})$$

If $\epsilon < \bar{h}$, the score bound increases with C ; the coverage upper bound is non-increasing in C .

<35%
Lower risk

35–45%
Monitor

>45%
Check factors

MAIN RESULT

SHAP Concentration Predicts Coverage Drop

Primary endpoint: 16 multiclass tasks across 9 domains

Scope: GBM multiclass classifiers, $K \geq 4$

$\rho = 0.853$
Spearman Correlation
 $p < 0.001$ | $n = 16$ tasks | 9 domains

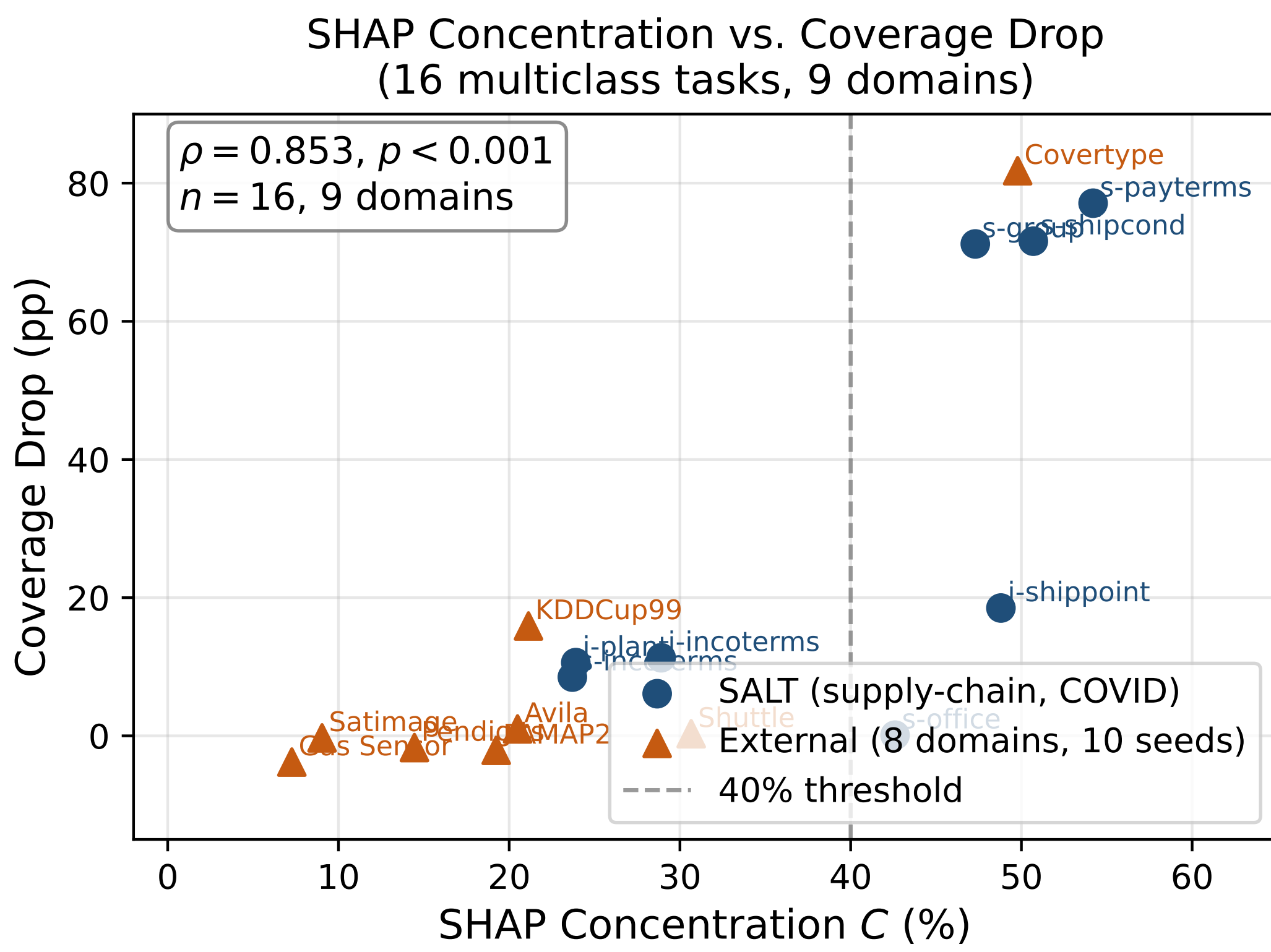


Fig 3: Each point is a task; x-axis = pre-deployment SHAP concentration, y-axis = post-shift coverage drop.

Takeaway: higher concentration tracks larger CP coverage loss.

Across 16 tasks, high- C models occupy the high-drop region.

External Specificity

4 robust external checks, all $C < 40\%$

Distinct from the $n = 16$ UCI endpoint; tests specificity: low C should predict robust coverage.

Dataset	C	Drop	Pred.
CivilComments	26.2%	-0.7pp	Robust
Amazon	1.9%	+0.4pp	Robust
Covertypes	12.5%	+8.0pp	Robust
Adult	27.5%	+1.4pp	Robust

91.7%

11/12 tasks correctly flagged
task-level robust/at-risk threshold rule

Next Steps

Future directions

- Gradient-based explanations for neural networks
- Automatic mitigation to reduce concentration
- Tighter distribution-specific coverage bounds

Open questions

- Can models be intrinsically **low-concentration**?
- Does the cutoff hold prospectively across domains?
- Can streaming CP trigger mitigation before failure?

Mechanism: Concentrated Dependence

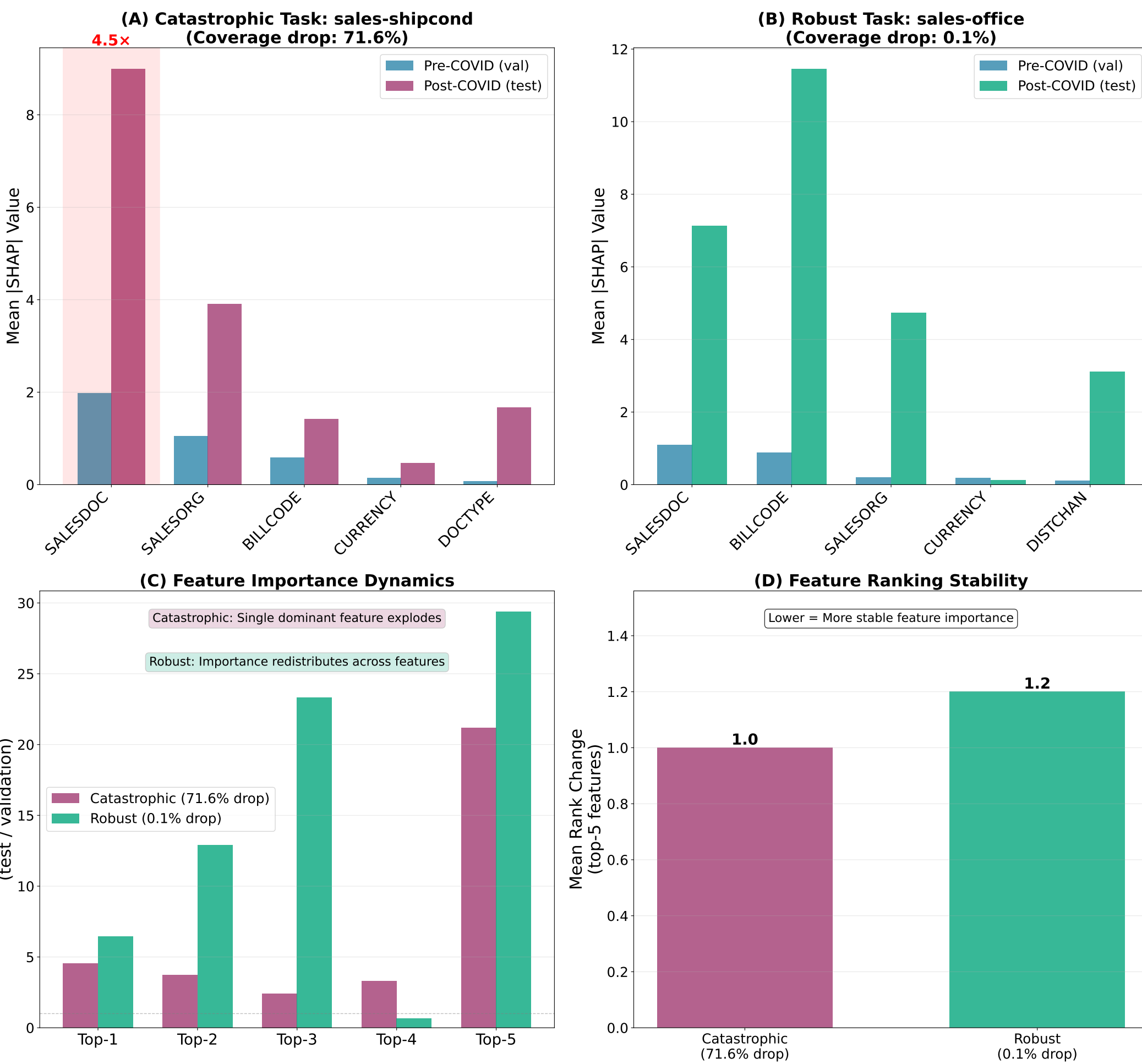


Fig 2: Mechanism. A/B compare SHAP mass before vs. after COVID shift; C/D show whether the failure is driven by one feature or distributed dependence.

- A. Failure:** post-COVID SHAP mass jumps onto one feature.
- B. Robust:** importance remains spread across features.
- C. Ratio:** failing task has a top-feature spike.
- D. Stability:** robust dependence changes less; the top features keep similar order after shift.

Takeaway: high- C GBMs create a single point of failure.

When the dominant feature shifts, APS scores rise and CP coverage drops.

Paper, Code & Context

Paper & Code

github.com/ChorokLeeDev/
conformal-covid-uai2026
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Scan for paper & code

Why Not Standard Shift Tests?

MMD, C2ST, and PSI detect shift across SALT tasks, but correlate weakly with coverage drop: $\rho \leq 0.19$. They say “shift happened,” not which CP model fails.

Scope & Limits

Best supported for **GBM multiclass classifiers**. Treat **35–45%** as uncertain; RF and MLP may show different failure modes.